

Question 1:

(a)

$$-\frac{V_{in}}{RC} = -5.6 \frac{V}{ms} \Rightarrow V_{in} = 5.6 \times 2.5 \times 10^3 \times 2 \times 10^{-6} \times \frac{1}{10^{-3}} = 28 V$$

$$\frac{V_{ref}}{RC} = 8 \frac{V}{ms} \Rightarrow V_{in} = 8 \times 2.5 \times 10^3 \times 2 \times 10^{-6} \times \frac{1}{10^{-3}} = 40 V$$

(b)

$$t = t_1 + t_2 = t_1 \left(1 + \frac{V_{in}}{V_{ref}}\right)$$

$$\text{Now, } V_1 = -56 = -\frac{V_{in}}{RC} t_1 \Rightarrow t_1 = 10 ms$$

$$\therefore t = 17 ms$$

$$N = \frac{V_{in}}{V_{ref}} 2^4 - 1 = 10.2. \text{ So, the binary output is } (1010)_2.$$

(c)

$$V_{out} = \frac{R_f}{2R} V_{ref} (1 + 0 + 0.25 + 0) = 25 V$$

(d)

$$\Delta = \frac{R_f}{2R} \frac{V_{ref}}{2^3} = 2.5 V$$

$$V_{out_{max}} = \frac{R_f}{2R} V_{ref} (1 + 0.5 + 0.25 + 0.125) = 37.5 V$$

$$(e) \text{ Error} = \frac{V_{in} - V_{out}}{V_{in}} \times 100 = 10.71\%$$

Question 2:

(a)

$$\Delta_{ADC} = \frac{16 - 0}{2^3} = 2 V$$

$$\Delta_{DAC} = \frac{R_f}{R} \frac{V_{ref}}{2^2} = 1 V$$

(b) (i) 4 V — 6 V

(ii) Lowest output from the DAC = 0 V. Step size = 1 V. Thus, outputs within the range in (i) are 4, 5 & 6 V. They are the outputs of '100', '101' & '110' respectively.

(c)

$$R'_f = 2R_f = 20 k\Omega$$

$$\Delta'_{DAC} = \frac{204}{104} = 2V$$

(d) $n_R = 2^2 = 4$, $n_{OA} = 2^2 - 1 = 3$. Encoder: 4 to 2.

Question 3:

(a) $V_{TH} = \frac{2}{2+3}15 = 6V$, $V_{TL} = \frac{2}{2+3}(-15) = -6V$

$$T_1 = R_x C_x \ln \left(\frac{-6 - 15}{6 - 15} \right) = R_x C_x \ln(6) \Rightarrow \frac{1}{2 \times 500} = 22 \times 10^3 \ln(6) \times C_x \Rightarrow C_x = 53.65 nF$$

$$\tau = R_x C_x = 1.18 ms$$

(b)

$$T_1 = R_x C_x \ln \left(\frac{V_{TL} - V_H}{V_{TH} - V_H} \right) = R_x C_x \ln \left(\frac{\frac{R_1}{R_1 + R_2} V_L - V_H}{\frac{R_1}{R_1 + R_2} V_H - V_H} \right)$$

$$= R_x C_x \ln \left(\frac{\frac{R_1}{R_1 + R_2} (-V_H) - V_H}{\left(\frac{R_1}{R_1 + R_2} - 1 \right) V_H} \right) = R_x C_x \ln \left(\frac{- \left(\frac{R_1}{R_1 + R_2} + 1 \right)}{\frac{R_1}{R_1 + R_2} - 1} \right) = T_2$$

$$T = 2T_1 = 2R_x C_x \ln \left(\frac{2R_1 + R_2}{R_2} \right)$$

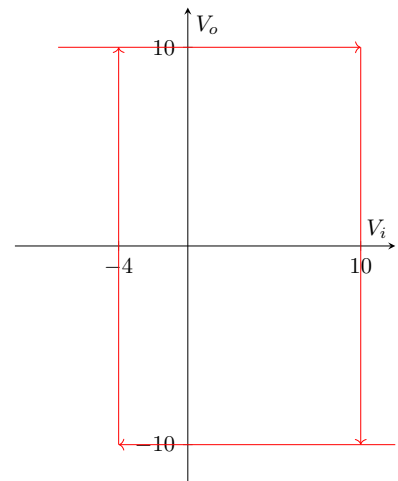
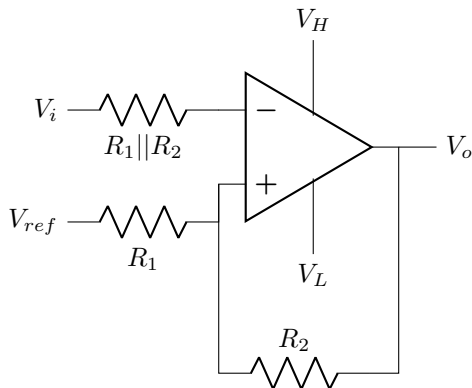
$$\Rightarrow T_{new} = 2R_x C_x \ln \left(2 \frac{R_1}{R_2} + 1 \right)$$

$$\Rightarrow \frac{1}{f_{new}} = \frac{1}{2f} = \frac{1}{2 \times 500} = 2 \times 1.18 \times 10^{-3} \ln \left(2 \frac{R_1}{R_2} + 1 \right)$$

$$\Rightarrow \frac{R_1}{R_2} = 0.26$$

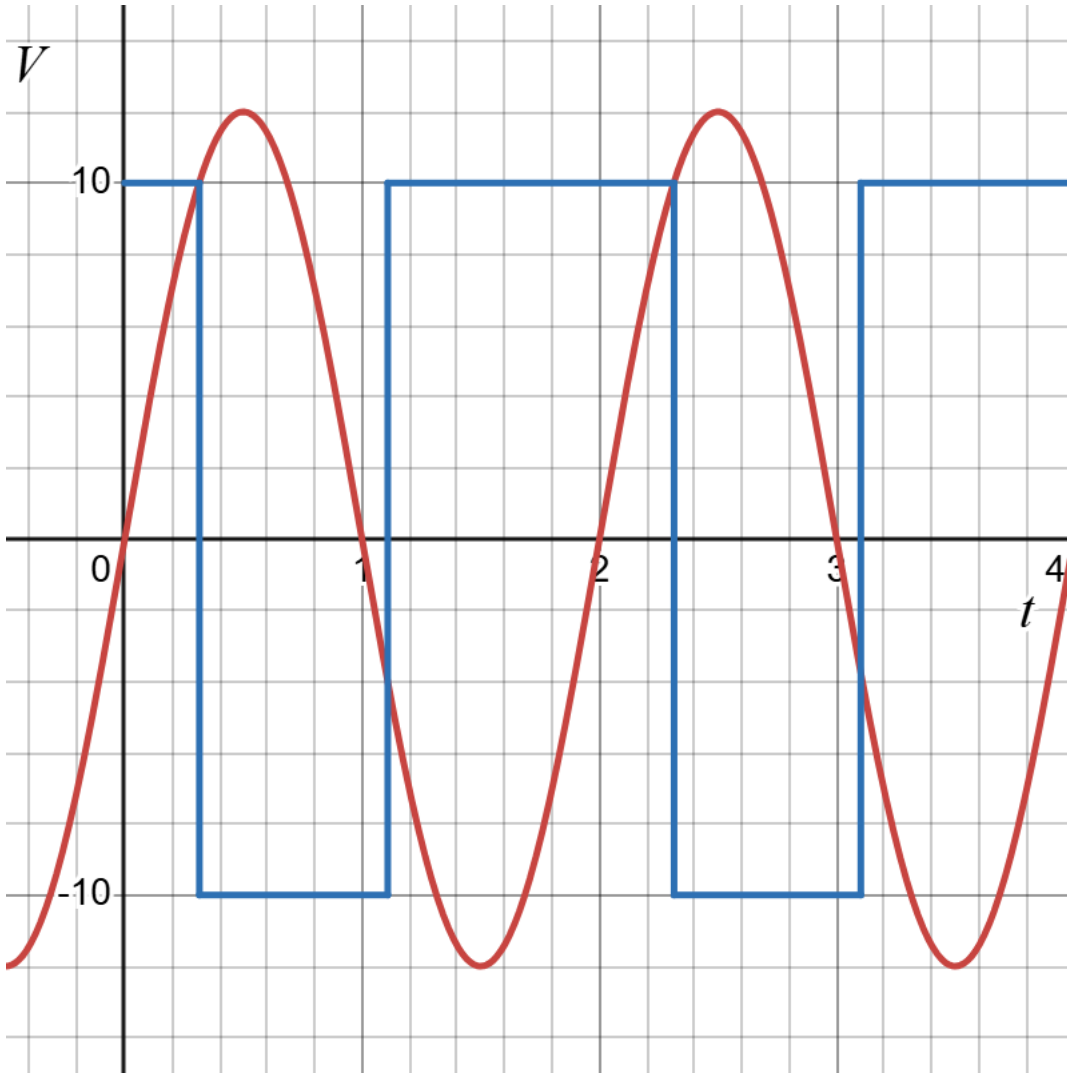
(c) Inverting. $V_H = 10V$, $V_L = -10V$, $V_{TH} = 7V$, $V_{TL} = -7V$, $V_S = 0V$, $V_{HW} = 14V$

(d)



$$R_1 = 7\text{ k}\Omega, R_2 = 3\text{ k}\Omega, V_{ref} = \frac{3}{7+3} = 10\text{ V}, V_H = 10\text{ V}, V_L = -10\text{ V}$$

(e)



Question 4:

(a) $T = 6\text{ ms}$

$$V_{TH} = 4 = -\frac{R_1}{R_2}(-8) \Rightarrow \frac{R_1}{R_2} = \frac{1}{2}$$

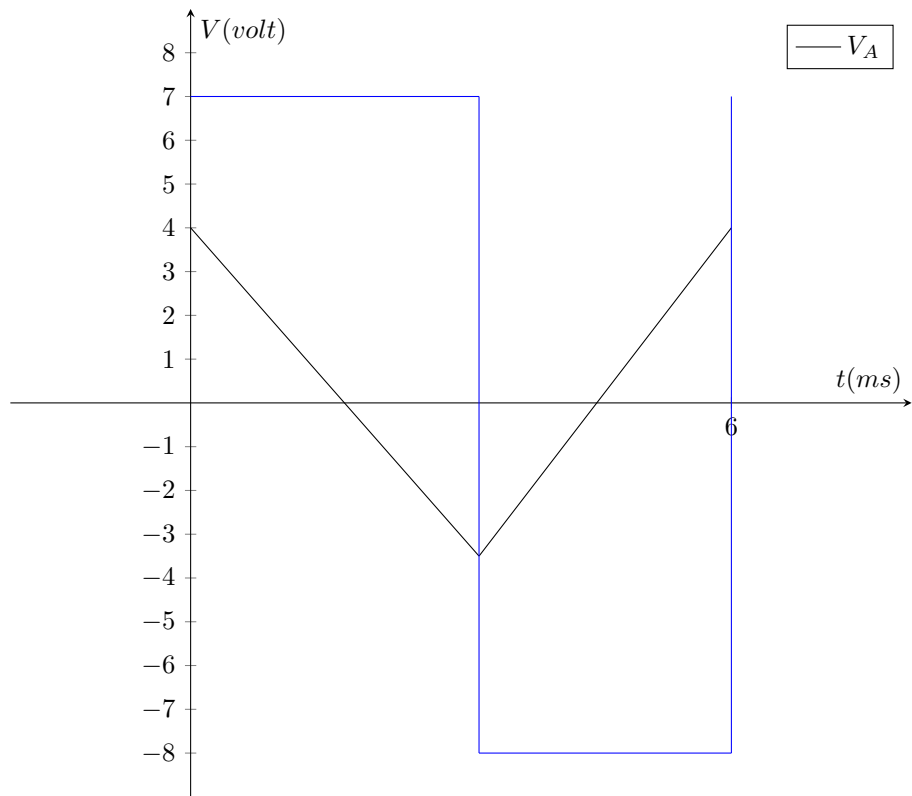
$$V_{TL} = -\frac{1}{2}7 = -3.5\text{ V}$$

$$\therefore D = \frac{T_2}{T_1 + T_2} = \frac{\frac{V_{TL} - V_{TH}}{V_L}}{\frac{V_{TH} - V_{TL}}{V_H} + \frac{V_{TL} - V_{TH}}{V_L}} = 46.67\%$$

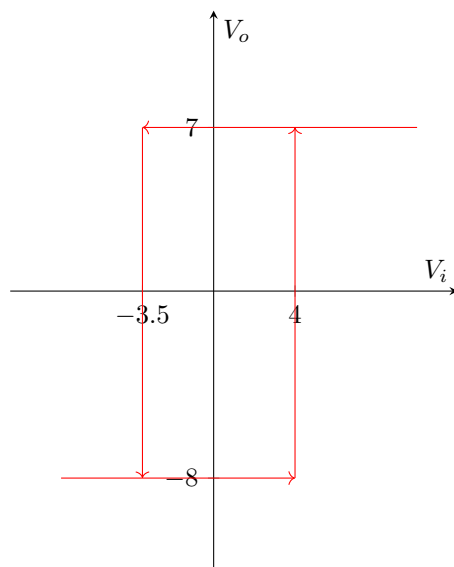
$$T_2 = D \times 6 \times 10^{-3} = 2.8\text{ ms}$$

$$C_i = \frac{T_2}{R_i \frac{V_{TL} - V_{TH}}{V_L}} = 1.49\text{ }\mu\text{F}$$

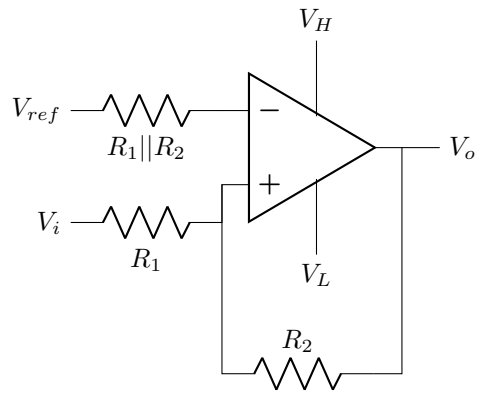
(b) $D_{sqr} = (1 - D_{tri}) \times 100 = 53.33\%$



(c)



(d)



$$V_H = 7V, V_L = -8V, R_1 = 1k\Omega, R_2 = 2k\Omega, V_{ref} = \frac{2}{\frac{1+2}{2}} = 1.33V$$

